This post is an attempt to explain the arc of a mathematical statistics course in a way that still feels useful outside the course itself.

It is inspired by the first half of Data 145, but I am rewriting it for a broader audience: someone who wants to understand how ideas like MLE, Fisher information, bootstrap, and Neyman-Pearson actually connect.

The goal is not to reproduce every proof or every computational detail from lecture. The goal is to keep the main flow, the core formulas, and the conceptual turns that make the subject hang together.

At the end, I also add a short bridge to reward-based post-training in modern AI, because some of the same ideas reappear there in a different vocabulary: KL penalties, proxy objectives, and exploitation of imperfect rewards.

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1. The big picture of the course

The course has a very clean progression once the lectures are connected.

1. Start with a real statistical question. We want to estimate something meaningful from data, not just compute a formula in the abstract.

2. Use a model to reduce complexity. In Lecture 1, the Poisson process turns a hard “full distribution” problem into a one-parameter problem.

3. Estimate the parameter. This naturally leads to maximum likelihood estimation (MLE).

4. Ask what the estimator does across repeated samples. Is it consistent? Approximately normal? Efficient?

5. Build general tools for that analysis. This is where convergence in distribution, Slutsky, and the delta method come in.

6. Generalize from one example to a full theory of estimation. Statistical models, plug-in estimators, score functions, Fisher information, asymptotic normality, Cramer-Rao.

7. Then complicate the story on purpose. MLE is not always the best estimator under the criterion we care about. This leads to decision theory, shrinkage, admissibility, and Bayes estimators.

8. Then complicate it again. Priors are not directly checkable. Models may be misspecified. Variances may be hard to compute analytically. This motivates objective Bayes, robust variance ideas, and bootstrap methods.

9. Finally, shift from estimation to testing. First test whole distributions using KS. Then move to the formal hypothesis testing framework and Neyman-Pearson theory.

In one line, the arc goes from a real question, to a model, to an estimator, to its sampling distribution, then to comparing estimators, then to Bayes / robustness / bootstrap, then to goodness of fit, and finally to hypothesis testing.

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2. What I am intentionally leaving in the background

There are a few topics I am deliberately not expanding in detail here:

- minimax calculations
- hierarchical Bayes and Gibbs sampling details
- the exact sandwich variance formula under misspecification
- randomized tests in discrete Neyman-Pearson settings

These ideas matter, but they are not the core of the story I want this post to tell.

What I do want to keep is why they appear:

- minimax because “best” depends on the comparison criterion
- hierarchical Bayes because priors can themselves be learned
- sandwich variance because wrong models change uncertainty calculations
- randomized tests because exact level constraints can be awkward in discrete spaces

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3. Lecture 1: the motivating story

Lecture 1 is not just an intro lecture. It quietly sets up almost the whole first half of the course.

The question

How likely is a significant earthquake in California within the next 7 days?

The key modeling step is to look at interarrival times between earthquakes.

The model

The lecture argues that:

- the interarrival histogram looks right-skewed, like an exponential distribution
- the cumulative count over time is roughly linear, suggesting a roughly constant event rate

That motivates a homogeneous Poisson process with rate λ.

If the process is Poisson with rate λ, then the interarrival times

$X_1, X_2, \ldots, X_{(n)}$

are i.i.d. exponential(λ).

Why this matters

This is the first big modeling lesson:

> A good parametric model can reduce an infinite-dimensional problem to a finite-dimensional one.

Instead of estimating the whole waiting-time distribution directly, we estimate a single parameter λ.

MLE appears immediately

For i.i.d. exponential(λ) data, the MLE is

$$
\hat\lambda_{MLE} = \frac{1}{\bar X_n}.
$$

This is already interesting because it is not just a sample mean. It is a nonlinear function of a sample mean.

The probability we actually care about

If $X \sim \text{Exponential}(\lambda)$, then

$\mathbb{P}(X\le 7) = 1 - e^{-7\lambda}$.

So the MLE-based plug-in estimator is

$\hat p_{(MLE)} = 1 - e^{-7\hat\lambda}$.

The lecture also compares this to a more direct empirical estimator based on the proportion of observed waits below 7 days.

The key lesson

Both estimators are approximately Gaussian, but the MLE-based estimator has lower variance if the model is right.

This is the first major payoff of parametric statistics:

> If the model is reasonable, structure can buy you efficiency.

Why this lecture leads directly to Lecture 2

The lecture asks:

- Why is $\hat\lambda = 1/\bar X_{(n)}$ approximately normal?
- More generally, why should a smooth function of an approximately normal quantity still be approximately normal?

That is exactly the setup for the delta method.

Key takeaways from Lecture 1

- Poisson process ⇒ exponential interarrival times
- why the earthquake story motivates a parametric model
- $\hat\lambda_{(MLE)} = 1/\bar X_{(n)}$
- plug-in estimation of $1 - e^{-7\lambda}$
- why Lecture 1 motivates CLT + delta method rather than proving them

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4. Lecture 2: convergence and the delta method

Lecture 2 is the mathematical bridge that turns Lecture 1’s intuition into something usable.

The main objects

The lecture reviews:

- convergence in distribution
- convergence in probability
- continuous mapping
- Slutsky's theorem
- delta method

The delta method

The version to remember is:

If

$$
\sqrt{n}(Y_n - \theta) \xrightarrow{d} N(0,\sigma^2)
$$

and g is differentiable with $g'(\theta) \neq 0$, then

$$
\sqrt{n}(g(Y_n) - g(\theta))
\xrightarrow{d}
N\left(0, (g'(\theta))^2 \sigma^2\right).
$$

Why it matters

This is one of the core moves of the class:

> Once I understand the asymptotic distribution of an estimator $\hat{\theta}$, I also understand the asymptotic distribution of smooth functions $g(\hat{\theta})$.

That matters all over the place:

- Lecture 1: $\hat{\lambda} = 1/\bar{X}_{(n)}$
- plug-in estimators in Lecture 3
- transformed parameters and probabilities later on

Intuition

The reason it works is local linearization:

$g(Y_{(n)}) \approx g(\theta) + g'(\theta)(Y_{(n)}-\theta)$.

So asymptotically, $g(Y_{(n)})$ behaves like a constant plus a scaled version of $Y_{(n)}$.

Key takeaways from Lecture 2

- the difference between convergence in distribution and convergence in probability
- what continuous mapping and Slutsky let me do
- the exact statement of the delta method
- how to use the delta method to move from $\hat{\theta}$ to $g(\hat{\theta})$

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5. Lecture 3: models, estimators, likelihood

Lecture 3 zooms out. Instead of focusing on one earthquake example, it builds the general language of statistics.

Probability vs. statistics

The course keeps emphasizing this switch:

- probability: distribution known, study random outcomes
- statistics: data observed, distribution unknown, infer structure

That shift in perspective drives everything else.

Parametric vs. nonparametric

A parametric model assumes the data come from a family

$f_{(\theta)}, \quad \theta \in \Theta$,

where θ is finite-dimensional.

A nonparametric model does not assume a fixed finite-dimensional form.

This distinction becomes important again later:

- parametric MLE theory
- nonparametric bootstrap
- KS testing of full distributions

Estimators, consistency, asymptotic normality

Lecture 3 sets up the three recurring estimator questions:

1. Is the estimator close to the truth for large n?
2. What is its approximate sampling distribution?
3. What happens if I transform it?

That leads to:

- consistency
- asymptotic normality
- plug-in estimators
- delta method for transformed estimators

Plug-in estimation

If I care about $g(\theta)$, the natural estimator is

$g(\hat{\theta})$.

This is simple but conceptually huge. A lot of the course can be summarized as:

1. analyze $\hat{\theta}$
2. transfer that result to $g(\hat{\theta})$

Likelihood and why MLE makes sense

For i.i.d. data, the likelihood is

$$
\text{Lik}(\theta;X) = \prod_{i=1}^n f_\theta(X_i),
$$

and the log-likelihood is

$$
\ell_n(\theta;X) = \sum_{i=1}^n \log f_\theta(X_i).
$$

The MLE is the θ that maximizes this.

Consistency of the MLE

The lecture's main idea is:

- by the law of large numbers,

$$
\frac{1}{n}\ell_n(\theta) \approx \mathbb{E}_{\theta_0}[\log f_\theta(X)]
$$

- the expected log-likelihood is maximized at the true parameter θ_0
- so the empirical maximizer should be near the true maximizer for large n

This gives the heuristic for MLE consistency.

Score function

The score is the derivative of the log-likelihood:

$S_{(n)}(\theta) = \ell_{(n)}'(\theta)$.

At the true parameter, its expectation is zero: